Linear Regression using Least Squares method Solved Example

- The following table shows the Mid Term and Final Exam grades obtained by the students in database course.
- Use the method of Least Squares to find the equation for the prediction of student's final exam grade based on the student's midterm grade in the course.
- Predict the final exam grade of a student who received an 86 in the midterm exam.

X Mid Term	Y Final Exam
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

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- Here, X is independent variable and y is dependent variable.
- Equation of linear regression looks like:

$$y = a + bx$$

- $\boldsymbol{a}$ is the intercept and $\boldsymbol{b}$ is the slope or coefficient
- $\boldsymbol{b} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$
- $\bar{\boldsymbol{y}} = \boldsymbol{a} + \boldsymbol{b}\bar{\boldsymbol{x}}$

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X Mid Term	Y Final Exam	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})*(y-\bar{y})$	$(x-\bar{x})^2$
72	84	-0.17	10	-1.7	0.03
50	63	-22.17	-11	243.87	491.51
81	77	8.83	3	26.49	77.97
74	78	1.83	4	7.32	3.35
94	90	21.83	16	349.28	476.55
86	75	13.83	1	13.83	191.27
59	49	-13.17	-25	329.25	173.45
83	79	10.83	5	54.15	117.29
65	77	-7.17	3	-21.51	51.41
33	52	-39.17	-22	861.74	1534.29
88	74	15.83	0	0	250.59
81	90	8.83	16	141.28	77.97
$\bar{x} = 72.17$	$\bar{y} = 74$			2004	3445.67

$$b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$$

$$b = \frac{2004}{3445.67}$$

$$b = 0.5816$$

$$\bar{y} = a + b\bar{x}$$

$$a = \bar{y} - b\bar{x}$$

$$a = 74 - 0.5816 * 72.17$$

$$a = 32.02$$

$$y = a + bx$$

$$y = 32.02 + 0.5816x$$

$$y = 82.04$$

Value of mid term grades ex: if mid exam grades are 86 then final 82.04